CLASS 9
1.2.4- MODELLING

Q1. What are the various stages of the AI Project Cycle?

- Problem Scoping – Identifying and defining the problem to be solved
- Data Acquisition – Collecting relevant data from reliable sources
- Data Exploration – Visualising data to find trends and patterns
- Modelling – Building AI models (Rule-based or Learning-based) using the data
- Evaluation – Testing the model's accuracy and performance

Example: *For a weather-prediction app – scope the rainfall problem, collect weather data, explore trends, build a model, then evaluate its predictions.*

Q2. What is Artificial Intelligence? Give an example.

Artificial Intelligence (AI) is any technique that enables computers to mimic human intelligence, working on algorithms and data to give desired outputs.

Example: *Voice assistants like Google Assistant or Alexa understand our speech and respond intelligently, helping us set reminders, play music, or answer questions in daily life.*

Q3. How is Machine Learning related to Artificial Intelligence?

Machine Learning (ML) is a subset of AI. While AI is the broader concept of machines mimicking human intelligence, ML specifically enables machines to improve at tasks through experience by learning from data and correcting past mistakes, rather than just following fixed instructions.

Q4. Compare and contrast Rule-based and Learning-based approach in AI modelling.

Basis	Rule-Based Approach	Learning-Based Approach
How it works	Machine follows fixed rules given by the developer	Machine learns patterns itself from labelled data
Adaptability	Static – doesn't change with new data	Adaptive – adjusts to new/unseen data
Data need	Works with limited, well-defined data	Needs large amounts of data to learn well
Example	Deciding to play golf based on fixed weather rules	Identifying apples vs bananas from new images
When to use	When rules are clear and unlikely to change	When data keeps changing or is too complex for fixed rules